
TECHNICAL ASSESSMENT PRESENTATION

CareLink National EHR

Architecture, Delivery
& Resilience

EHR

NATIONAL HEALTH RECORD

Continuity
Integrity
Scale

CareLink must work where connectivity does not

1,500

FACILITIES

National footprint

~3,000

CONCURRENT USERS

Weekday-morning peak

1,100

CONNECTIVITY CONSTRAINED

800 intermittent + 300 offline periods

DESIGN PRIORITIES

01 Clinical continuity

02 Data integrity

03 Security & privacy

04 Operational
simplicity

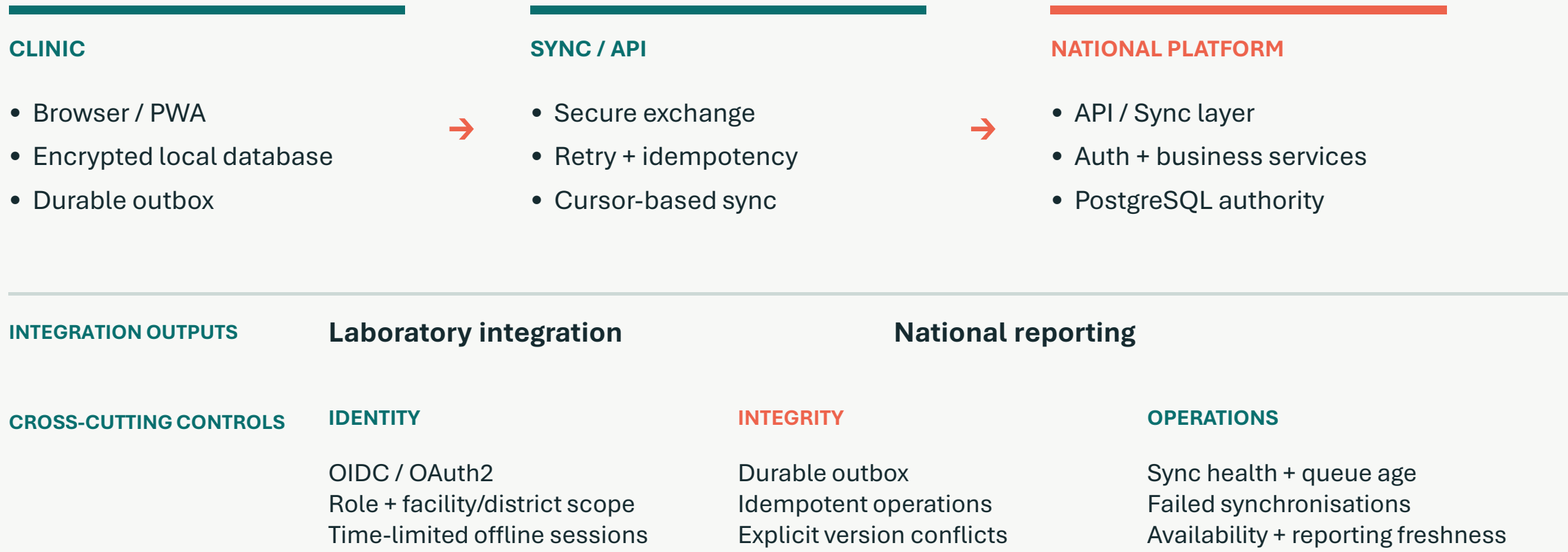
OPERATING CONSTRAINTS

Older clinic hardware and browsers
Sensitive patient information

ASSUMPTION

A visit before the missed appointment does not clear
follow-up; a visit after it does.

Offline-first clinics sync into one authoritative national record



Three choices reduce clinical and delivery risk

01**OFFLINE-FIRST**

**Local work + durable
synchronisation**

REJECTED

Online-only

WHY IT MATTERS

Connectivity cannot become a clinical dependency when 1,100 facilities cannot rely on continuous connectivity.

02**MODULAR MONOLITH FIRST**

**Well-separated modules in one
deployable platform**

REJECTED

Immediate microservices

WHY IT MATTERS

The scale does not justify distributed-system complexity on day one.

03**IDEMPOTENT MUTATIONS**

**Every mutation has an operation
ID**

REJECTED

Treat every retry as a new request

WHY IT MATTERS

A timeout does not tell us whether the server committed the original request.

Local retention remains an evidence-led decision

Current position

- Minimum necessary facility data
- Encrypted local database
- OS/device-protected encryption keys
- Configurable stale-data retention
- User revalidation when connectivity returns
- No national patient population cached locally

What would change it?

- Actual maximum offline duration
- Clinical workflows
- Ministry privacy/security requirements
- Device-management capability
- Data sensitivity and volume



PRINCIPLE

Do not keep sensitive data locally simply because storage is available.

The vertical slice proves the workflow — not production readiness

BACKEND

- Django + Django REST Framework
- PostgreSQL
- Typed API responses
- Role/facility authorisation
- Server-side pagination
- Filtering and sorting
- Correlation IDs + structured errors
- Transactional contact recording

FRONTEND

- React + TypeScript
- API service layer
- Facility/status filters
- Days-overdue sorting
- Pagination
- Loading / empty / error / retry
- Accessible semantic controls
- Contact workflow

API + EVIDENCE

- GET /api/follow-up
- POST /api/follow-up/{id}/contacted
- 13 backend tests passing
- Frontend lint ✓
- Production build ✓

STATUS

WORKING

Core follow-up
vertical slice

DESIGNED NEXT

01 AUTH

OIDC / OAuth2

02 OFFLINE

Encrypted store +
outbox

03 WRITES

Idempotency + retry

04 SYNC

Versioned conflicts

05 INTEGRATE

Lab + reporting
boundaries

06 OPS

In-country deploy +
monitoring

Security and clinical integrity block merge

CRITICAL · SECURITY

- Hard-coded database credentials
- SQL injection
- Missing authorisation
- PHI / National ID in logs

CRITICAL · DATA INTEGRITY

- Swallowed exceptions
- Non-transactional writes
- Unsafe SQL writes
- Incorrect follow-up evaluation

MAJOR · SCALE & UX

- N+1 database queries
- No pagination
- 3-second frontend polling
- Client-side filtering
- Direct state mutation
- Non-semantic controls

THREE MERGE BLOCKERS

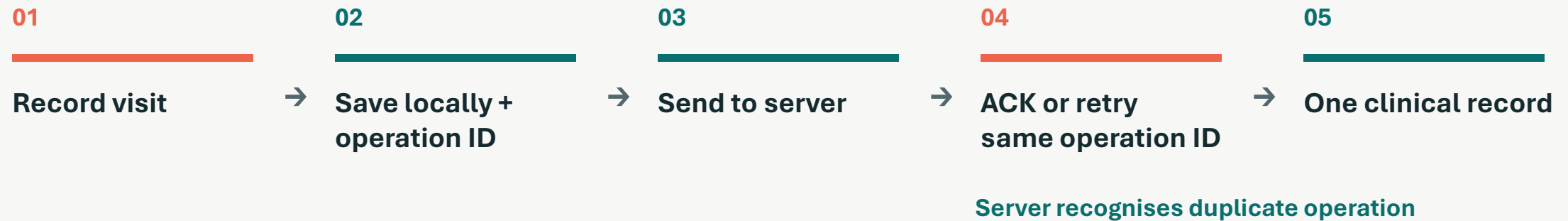
1 Security & authorisation

2 Clinical data integrity

3 Scale & follow-up correctness

Idempotency turns retries into exactly-once clinical effects

A timeout is not the same as a failure.



VERSION CONFLICTS

Clinic v42 → National v43
Explicit conflict → authorised review → decision recorded

DUPLICATE RECORDS

Detect → Preserve → Review → Merge / Canonicalise
Never casually delete clinical records.

Next week turns a working slice into production evidence

01 DAYS 1–2

- Harden authentication + authorisation
- Frontend/API automated tests
- Accessibility review
- API contract validation

02 DAYS 3–5

- Durable client outbox
- Idempotency store
- Retry / backoff
- Cursor-based synchronisation
- Conflict UX

03 DAYS 6–7

- 10k+ follow-up performance test
- Concurrent-user/load validation
- Offline/replay testing
- Security checklist
- Deployment + rollback rehearsal

GOAL

Not “more code” — evidence that the system is safe to scale.

Questions

Architecture · Delivery · Leadership

Continuity
Integrity
Scale